UNITED REPUBLIC OF TANZANIA

PRIME MINISTER'S OFFICE

ARUSHA REGION MOCK FORM TWO 2026

CHEMISTRY

MARKING SCHEME

SECTION A 15 MARKS

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	A	B	B	A	A	B	D	A	A

01@ = 10 marks

2

i	ii	iii	iv	v
E	H	G	B	C

01@ = 05 marks

SECTION B (70 MARKS)

3 a) i) MN_2 --- 01 mark

ii) Ionic / Electrovalent bond. 01½ mark

iii) - It is formed between a metal and non-metal atom

i - It is soluble in water
- It conducts electricity in solution or molten form
- It have high melting and boiling points
- It is crystalline solid at room temperature

any three points
01@ = 03 marks

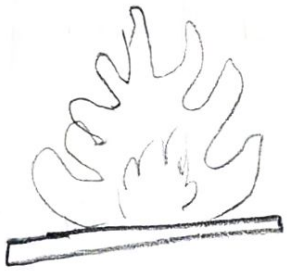
6)ii) A - Corrosive



ii) Container B - Toxic



iii) Container C - Flammable



01½@ = 04½ marks

4 a) i) Electrons 01 mark

ii)- They have high energy
- They have small mass

01½@ = 03marks

b) i) sublimation
ii) Layer separation
iii) Solvent extraction
iv) Fractional distillation
v) Chromatography
vi) Evaporation.

01@ = 06marks

5. a) i) Electronegativity is the tendency of an atom to attract shared electrons in a chemical bond towards itself

02@ = 04 marks

ii) Ionization energy is the the energy required to remove an electron from an atom or ion in its gaseous state.

b) i)

I							VIII
A	II	III	IV	V	VI	VII	
						B	C
			D				
E	F						

$0\frac{1}{2}$ = 03 marks

ii) F

iii) C

iv) A

v) B

vi) E

vii) D $0\frac{1}{2}$ @ = 03 marks.

6. a) i) Group V period 3 - 01 mark

ii) 15 electrons ---- 01 mark

iii) 16 neutrons ---- 01 mark

iv) Element X is a non-metal because it gains three electrons to form ion ---- 02 marks

v) $^{31}_{15}X$ 01 mark

b) $R.A.M = \frac{(P_1 \times A_1) + (P_2 \times A_2) + (P_3 \times A_3)}{100}$ --01mark

$= \frac{(90.5 \times 20) + (0.3 \times 21) + (9.2 \times 22)}{100}$ - 01mark

$= \frac{1810 + 6.3 + 202.4}{100}$

$= \frac{2018.7}{100}$

$= 20.187$ 02 marks.

$\underline{=} 20.2$

7.a) Oxidation is the loosing of electrons while Reduction is the gaining of electrons (02 marks)

b) i) It takes short time to write

ii) It can be balanced

iii) It occupy a small space

iv) It shows the proportion of atoms

v) It shows the physical states of the reactants and products Any four points 01@ = 04 marks

c) i) Decomposition

ii) Combination

iii) Displacement 01@ = 04 marks

iv) Precipitation / double decomposition.

8. a) i) Rusting of iron
ii) Rotting of fruits
iii) Decaying of meat
iv) Souring of milk
v) Digestation of food in the stomach
vi) Burning of fuels Any three 01@ = 03 marks

b) i) Its composition is fixed.
ii) It can not be separated by physical means
iii) Its formation involves chemical change
iv) Its properties are not those of its components
v) It have a specific formula
Any four 01@ = 04 marks

c) i) Lactic acid
ii) Formic acid
iii) oxalic acid
iv) Formic acid
v) Citric acid
vi) Tartaric acid
06 1/2 @ = 03 marks

9. a) i) Fires
ii) Falling down
iii) Electric shock
iv) Burns
v) Suffocation
vi) Cuts
vii) Damage of equipment
viii) Explosion
Any three 01@ = 03 Marks

b) i) It should have water supply
ii) It should be well ventilated
iii) It should have a fume chamber
iv) It should have fire extinguisher
v) It should have a first aid kit
Any three 01@ = 03 marks

c) i) Do not eat or drink in the laboratory
ii) Do not taste or smell chemicals
iii) Do not spill chemicals on the floor
iv) Do not touch an electrical appliance with wet hands
v) Do not use dirt, cracked or broken apparatus

Any four 01@ = 04 marks

SECTION C (15 marks)

10. a) i) Empirical formula is the formula which represent the simplest ratios of the atoms or ions in a compound. 02@ = 04 marks

ii) Molecular formula – is the formula which show the actual number of each kind of atoms in a molecule.

b) i) Na_2O

ii) $Ca(NO_3)_2$

iii) NH_4Cl

iv) KCN 01@ = 04 marks

c)(i)

Symbols	N	O
% composition	30.4	69.6
R.A.M	14	16

00½ mark

Divide % composition by R.A.M	$\frac{30.4}{14} = 2.17$	$\frac{69.6}{16} = 4.35$ 00 1/2 mark
Divide by the smallest value	$\frac{2.17}{2.17} = 1$	$\frac{4.32}{2.17} = 2.004$ 00 1/2 mark
Ratios	1	2 00 1/2 mark

The empirical formula is NO_2 - 01 mark

ii) n(empirical formula) = molecular mass

molecular mass = vapour density x 2

= 46 x 2

= 92 ----- 01 mark

$n(NO_2) = 92$

$n(14+32) = 92$

$\frac{46n}{46} = \frac{92}{46}$

$n = 2$ ----- 01 mark

n(empirical formula) = molecular formula

$= 2(NO_2)$

$= N_2O_4$ ---- 01 mark

Molecular formula is N_2O_4

iii) Name for the compound is Dinitrogen tetraoxide

01 mark